

Existing Challenges and Failure Modes

Primary project source: ProofRoad v2, SIH 2026 PS 26007, uploaded by the user. The document describes ProofRoad as an engineering proposal and explicitly says it is not a safety certification. Page references below refer to the uploaded ProofRoad v2 document.

1. Environmental challenges

Fog, rain, humidity and dust degrade different sensing modalities differently. ProofRoad explicitly treats RGB and NIR as weak in dense fog, thermal as useful for warm objects but still range-limited, and radar as the primary ranging modality (pp. 3–4).

Mine haul roads also introduce clutter from pit walls, reflections, uneven surfaces and changing road geometry.

2. Human-machine challenges

The driver must continue watching the windshield. ProofRoad estimates that at 5 km/h a two-second glance at the display corresponds to about 3 m of travel, which is significant when visibility is only 3–5 m (p. 7).

Warning fatigue is addressed by making alerts speed-conditional, issuing one alert per hazard and repeating only on escalation (p. 8).

3. Communication and trust challenges

Radio loss cannot be allowed to become a false-clear condition. ProofRoad treats silence as uncertainty, lets peer information become stale, and keeps the local radar loop independent of the radio (pp. 3, 9–10).

Messages that add caution can be accepted with valid age/signature checks. Messages that relax restrictions require stronger local checks because a forged clear message is more dangerous than a forged caution message (p. 10).

4. Safety architecture challenges

The proposal separates a safety MCU from an assist computer. The MCU reads radar, runs the SafeCorridor formula, drives alerts and has a watchdog; the Pi/laptop handles RGB/thermal/NIR, naming and dashboard functions. If the assist computer fails, the radar warning remains available (p. 3).

ProofRoad explicitly states that the prototype is advisory and does not claim certification. Real intervention would require machine-maker integration and functional-safety work (pp. 13–14).

5. Evidence gaps that must remain explicit

The demo radar is a stand-in rather than mine-grade 77 GHz hardware.

Braking values are prototype measurements and must be replaced with truck-specific tested values.

Radio behavior in a real pit is unproven.

Fog-machine droplets are not equivalent to natural mine fog.

Only two physical trucks are available, so fleet performance is simulated.

Prototype security uses shared demo keys and has no penetration test.

Human-factors performance cannot be established in a hackathon.

6. Research/test matrix

Failure / uncertainty	Test	Pass evidence
Radar degradation	Target detection under fog/rain/dust	Range + miss/false-alarm measurements
Braking uncertainty	Repeated stopping tests with representative load/surface	Conservative braking envelope
Radio loss	Disable ESP-NOW/LoRa during run	Truck becomes more cautious; no false-clear
Camera/AI crash	Kill assist computer	MCU/radar alerts continue

Failure / uncertainty	Test	Pass evidence
Forged message	Replay and forged GO message	Rejected/logged; local limit unchanged
Gate uncertainty	Drop gate messages / sensor	Entry blocked; safe state maintained